Changing states

MATTER CHANGES FROM ONE STATE TO ANOTHER ACCORDING TO TEMPERATURE AND PRESSURE.

SEE ALSO
❮ 96–97 Properties of materials
❮ 98–99 States of matter
Convection currents 189 ❯

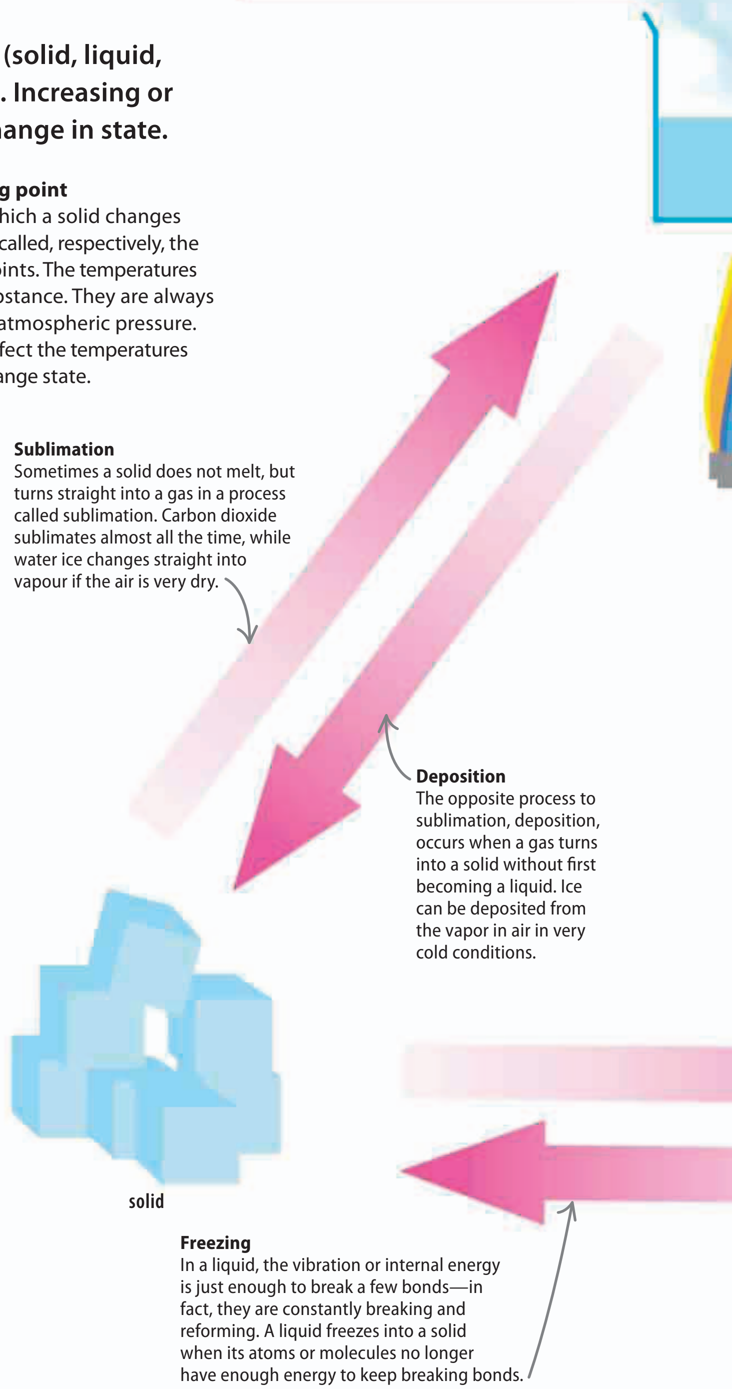
Every substance has a standard state. This is its state (solid, liquid, or gas) at 25°C (77°F)—just above room temperature. Increasing or decreasing that temperature eventually leads to a change in state.

States and energy

Changes in state are the result of energy being added to or removed from a substance. Taking energy from a gas results in it becoming a liquid and then a solid. Adding energy has the reverse effect. The energy within a substance makes its basic units—atoms or molecules—vibrate (wobble). This vibration, called internal energy, is measured when temperatures are recorded.

SEAgel is a spongelike solid made from seaweed. It is so light that it floats in thin air!

▷ Melting and boiling point
The temperatures at which a solid changes into a liquid or gas are called, respectively, the melting and boiling points. The temperatures are specific to each substance. They are always measured at standard atmospheric pressure. Changes in pressure affect the temperatures at which substances change state.



REAL WORLD

Salting ice

Adding salt to ice lowers the melting point of water by a couple of degrees. Salting roads in winter stops dangerous sheets of ice from forming—although if the conditions are well below 0°C (32°F) the water will still freeze. The salt dissolved in the water gets in the way of the water molecules, making it harder for them to form all the bonds they need to become ice.